Mobile + Web Based Knowledge Sharing

Social Media Platform

Higher Diploma in Software Engineering

Final Project Report

23.2F



School of Computing & Engineering

National Institute of Business Management

Galle

Mobile + Web Based Knowledge Sharing Social Media Platform

HIGHER DIPLOMA IN SOFTWARE ENGINEERING

FINAL PROJECT DOCUMANTATION

HDSE23.2F

Submitted By

GAHDSE23.2F-001 - G.K.A Rashini Shehana

GAHDSE23.2F-021 - K.K.C Imangana

Year of submission – 2025

The project is submitted in partial fulfilment of the requirement of the Higher Diploma of Computer System

Declaration

“We hereby, declare that the work is our original work and has not been submitted previously for a diploma or degree at this or any other university/institute. To the best of our knowledge, it does not contain any material published or written by another person, except as acknowledged in the text.”

…….. ……..

Project Supervisor Date

NIBM Galle.

Acknowledgment

All participants who contributed in the successful completion of our project receive our most profound appreciation. The completion of this project became possible because numerous people provided their guidance and encouragement throughout the entire process. Miss. Sandaruwani Pathirage, Miss.Thilini Darmasena and Mr. Viraj deserve our greatest thanks because they provided important guidance and unending support throughout our project. The supportive feedback and motivating words from them were central in developing our project and we appreciate the dedication along with the guidance that they provided. Fundamental appreciation goes to all lecturers who have continuously taught valuable knowledge to us throughout our academic phase. Through their guidance we developed essential abilities and received important revelations that were instrumental to our project achieving success. Our work reached its potential for excellence through their valuable ideas and meaningful feedback. Our sincerest thanks are directed toward all our parents and family members as well as our friends because of their persistent motivational support throughout the project. The strength we obtained for overcoming obstacles alongside keeping our focus on our goal stems from our parents' steadfast belief in us and their constant encouragement.

We are grateful to have received the support of many vital people who guided us through our project while consistently inspiring our efforts which led to its achievement.

Abstract

In the environment of swift software engineering, it is essential to have a platform that relates learning with experience. Here is the plan of the creation of a modern and innovative platform for knowledge sharing, a social media aimed at connecting students and resource people to improve the educational process. This application will be in the form of a mobile application that can be installed in various mobilities, as well as a web-based application and it will have social networking elements combined with effective tools that will help in education. The essence of students / resource people will be control of the profile, online chats in real time, work with courses and assignments, and an individual outlook for employment with the help of machine learning. Thus, the goal of our platform is to eliminate the issues related to the lack of integration of resources, the inability to get access to the industry materials, and the existence of various administrative problems. It will create synergy of the learners and experts, who will regularly give and receive knowledge. The ultimate objective that we have set for ourselves is to build a responsive system of knowledge creation and exchange as well as career management. By integration of new technology and use of the friendly interface we plan to make a great difference not only for the Software engineering educational system but also for the students' population in the world. Object Oriented approaches have been used in developing. Php, html, JavaScript ect has used to develop the website. Java has used to develop the mobile app. Firebase, MySQL have been used to create to databases. Any web browser such as Edge, Chrome, Firefox can be used to access the website, and any android version can be used to access the app in any environment such as windows or Linux.

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Chapter 1: Introduction

Digital technologies are making the educational environment rapidly evolve. Due to the enhanced use of online tools, education systems are shifting from conventional types of learning to more appropriate and engaging types of learning. This project seeks to benefit from this trend by creating a mobile and web-based knowledge sharing platform through the design of a particularly enhanced environment for software engineering students. The established objective of the proposed platform is to reduce the gap between theory and practice so that the learning process is smooth. Unlike the traditional e-learning systems that are just informative, where content has only an informative function and serves as a one-source text message, the proposed platform mentions social media activity including real-time chats, group discussions, team assignments, etc., to make the students and experts engage in two-way communication. Some of the features of the platform are machine learning which improves the platform by providing personalized career coaching and performance and interest data analytics. More than in many other fields, in software engineering it is vital for a professional to make learning a lifelong process.

This platform aims to develop a community of learners and resource persons where they will be able to share knowledge, experience and industry practice in real-time. By offering accessibility and interactivity together with career-related features, this endeavor's purpose is to reimagine how learning and preparing for future careers can happen.

Also, the designed platform is easy to scale and targeted for users from different parts of the world, which also makes it multilingual. This unifies the look of both the mobile wireframe and the web browser, which when accessed would work equally well. This characteristic is important in the current multi-technology world where learning should be possible at any time, at any place. We also hope to improve such aspects of software engineering education as the lack of a collective knowledge base, difficulties in finding a mentor, and the absence of individual career counseling. In accomplishing these, the platform does help in the academic accomplishment of the learners while at the same time ensuring that the students make the right choices regarding their careers, hence ensuring they gain employability in the future.

1.1 Problem Definition

Several challenges in software engineering education include:

- Fragmented Learning Resources:

  The unstructured approach to the availability of learning materials and industrial inputs gives the students a hard task in compiling their knowledge.

- Limited Access to Resource People:

  Education institutions and learners lack contacts to get industry exposure and hence the gap between theory and practice is evident.

- Career Path Uncertainty:

  There is a lack of direction given to students when identifying relevant software engineering specialties and, as a result, productivity declines tremendously.

- Administrative Inefficiencies:

  Management of the roles of users adds to the work of the administrators together with the complexity of ensuring that the platform is secure.

- Engagement Barriers:

  Reduced chances of body contact and communication between the students and the experts decreases peer to peer and content sharing by tutors.

- Lack of Predictive Insights:

  There is a lack of proper platforms to offer students career information, course performance, skill and career interests.

1.2 Proposed Solution

The project proposes a knowledge-sharing platform with two primary modules for two forms of client interface: a mobile application and a web application. Combined with each other, it is expected that these modules can improve the quality of education provided through providing new additional functions and services required for software engineering students, resource persons, and administrators.

1.2.1 The Key Features & Their Benefits

- **User Registration and Profile Management:**
    - Students and resource people can create their own unique profile.
    - In the profiles set, activities are organized, progress is monitored, and suggestions are given based on individual characteristics are provided.
- **Real-Time Chats and Group Discussions:**
    - Co-ordinate student and resource persons to ensure proper interaction.
    - Allow students to discuss in groups to foster learning and solve problems.
    - Offer features for private conversations to organize a niche for direct guidance.
- **Course and Assignment Management:**
    - As a result, resource persons are empowered to build courses, upload relevant materials, and develop assignments that meet learning outcomes.
    - Students can register for classes, submit work and view results.
    - It also has an org chart feature to assist clients in tracking their performance in their academics.
- **Career Path Predictions:**
    - Apply machine learning models to study the performance of students in their classes and the chosen product preferences.
    - Offer career suggestions relevant to the student, navigate him/her to the appropriate specialties.

- **Administrative Tools:**
    - It is easy for the administrators to maintain the users and course more so the system settings.
    - The features are built in the form of tools to track the activity on the platform and deal with security and user concerns.

1.2.2 How the Solution Measures Up to some of the Key Challenges

- **Fragmented Learning Resources:**
    - Coordinates educational material to generate a single source that will be easily accessible by anyone.
    - Extremely helpful when it comes to searching for courses, due and useful industry references.
- **Limited Access to Resource People:**
    - Bring students closer to employers and companies by arranging real time interaction and opportunities for mentoring.
    - Develop connections with skilled workers who may swap information and give career advice.
- **Engagement Barriers:**
    - Provides a means of better communication through options such as group and live forums.
    - Facilitates students learning from their colleagues and from experts from the same discipline.
- **Lack of Predictive Insights:**
    - Provides deeper analytics of the user utilizing the strengths and weaknesses to be determined to enable support.
    - It keeps records of usage activity in order to evaluate and enhance characteristics of the platform as well as customers' interactions.

1.3 Project Objectives

The major aims of this project are to meet the specific needs of software engineering education and to guarantee the effectiveness of the proposed platform to students, resource persons and administrators as well. Below is a detailed explanation of each objective:

- **Facilitate Seamless User Experience:**
  - Make both mobile and web site as clear as possible so that students and resource people can easily find their way around and through the site.
  - Try to use elements which will fully respond to different devices and different sizes of screens, this will help to make the website more available for people from the whole wide world.

- **Promote Collaborative Learning:**
  - Develop a platform by which students and resource persons can share ideas and solve problems as a team with an aim of trying to eliminate the theory practiced deficiency.
  - Promote actual time interaction and coordination by offering an environment where group talks, or group projects are achieved.

- **Empower Students:**
  - Thus, machine learning algorithms can be applied to calculate the student's performance and give an individual suggestion on which career might be more appropriate.
  - Facilitate students to join courses to meet particular learning objectives and improve the learning process.

- **Improve Interaction and Interactionnaires:**
  - Include components that enable students to talk to resource people while doing their tasks to have perpetual conversation on social networks.

- **Leverage Advanced Technologies:**
    - Engage appropriate analytics to monitor user's performance and interaction with the solution to update the platform constantly.
    - Integration of statistical forecast models to splendid decisions, for example, recognition of favorite courses or suggestion of the best study materials.
- **Streamline Administrative Tasks:**
    - Scholars support administrators with an effective means of handling user accounts, course contents, and system features.
    - Eliminate redundant administrative work, while at the same time increasing work efficiency and system dependability.

Chapter 2: Methodology

2.1 System Requirements

2.1.1 Functional Requirements of the System

- **Login Management Function**

  Designed to be used by students and resource people. This function provides the following functions:

  - Students and Resource People shall be able to sign in and sign up.
  - Students and Resource People shall be able to manage their profiles.

- **Resource People Function**

  Provides functionality for Resource People only. It will not be available to any other student of the system.

  - Resource People shall be able to create group chats.
  - Resource People shall be able to create chats.
  - Resource People shall be able to create Courses.
  - Resource People shall be able to add assignments.
  - Resource People shall be able to add students for courses.
  - Resource People shall be able to view submissions.
  - Resource People shall be able to upload Content Videos
  - Resource People shall be able to upload Subject Materials.
  - Resource People shall be able to View Reviews.

- **Student Function**

  This is the Simplest Module. Students can use this. It provides flowing functions:

  - Students shall be able to create chats.
  - Students shall be able to request courses.
  - Students shall be able to do assignments.
  - Students shall be able to view Score.
  - Students shall be able to view Content Videos.
  - Students shall be able to download Subject Materials.
  - Students shall be able to add Reviews for Courses.

- **User Management Function**

  Only useable for the Admin User.

  - Admin shall be able to add students and Resource People.
  - Admin shall be able to view/delete students and Resource People.

2.1.2 Non-Functional Requirements of the System

The non-functional requirements do not directly affect the system. But they are very special for success in the system. They Address safety and security, including different user levels with login credentials to control access to system functions based on their role.

These are the non – functional requirements for this system.

- The system should provide a user-friendly environment including a flexible interface.
- A person with average computer skills can work with the system with a short period of training.
- The system should be accurate and consistent when manipulating the fed data in the proper way and displaying correct information.
- The system should keep up security and reliability, because the system handles important data related to business processes of the company.
- Occasionally backups should be made to maintain reliability the system should be reusable and maintainable.

2.2 Proposed Technologies

2.2.1 Frontend Technologies:

- HTML: Employed to establish the structural system of the web-based application.
- CSS: Adds style and graphic layering to help make the overall design as easy and visually inviting as possible.
- JavaScript: Supports interaction and ability to update the content of the platform dynamically.

2.2.2 Backend Technologies:

- PHP: used to build the system.
- Firebase: Offers the immediate database dimension and identity services for safe and optimum data management.
- MySQL: Stands as the main organic database, which includes structured data and performs typing on the results to return only the data the application requires, such as user and course profiles and details about assignments..
- Java: Selected for mobile app development since it supports both Android and iOS and uses relatively few resources.

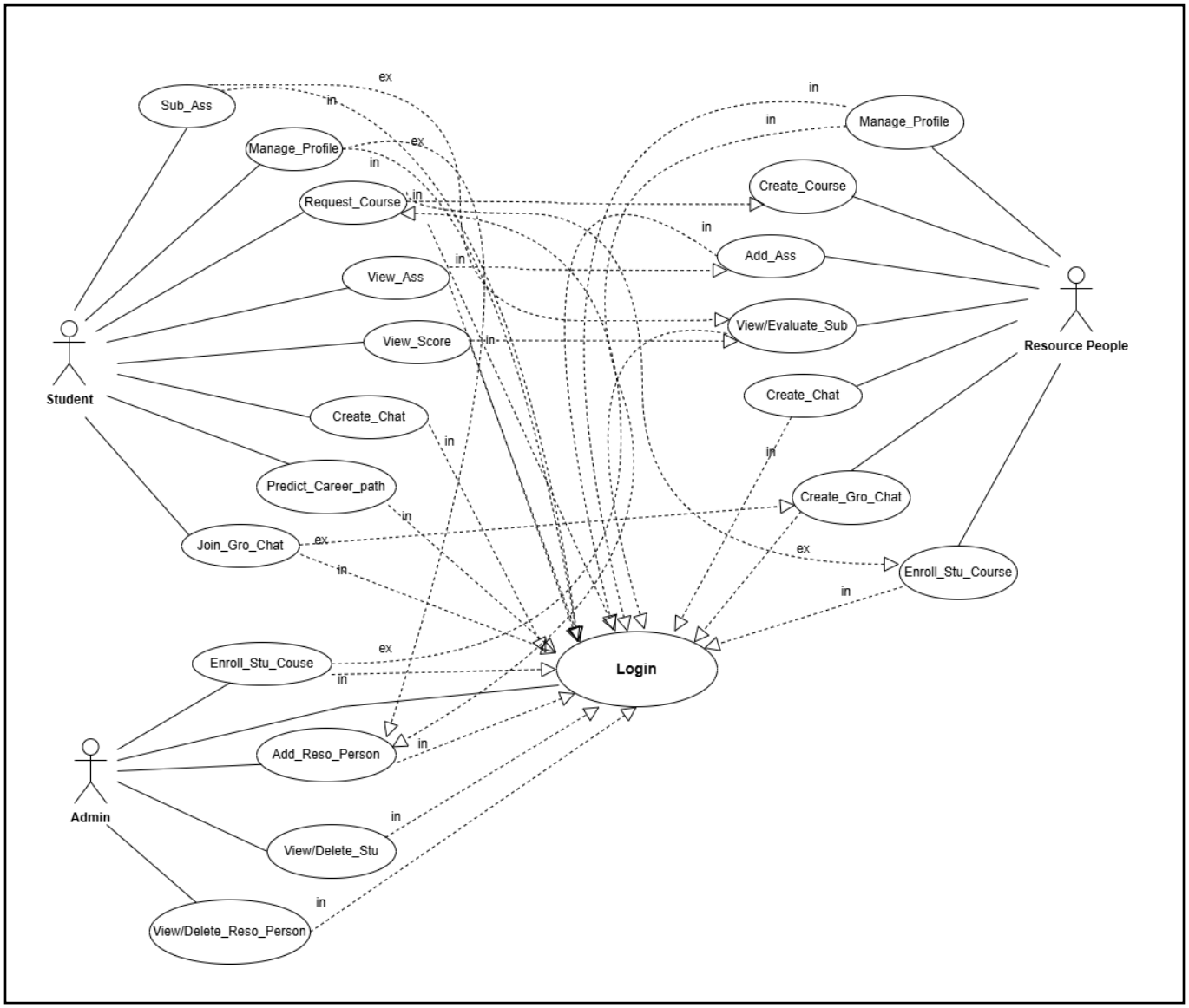
2.2.3 Development Tools:

- Android Studio: Helps in the development of Flutter applications through debugging and testing functionalities with a proper flow of the quickest growing mobile application.
- Sublime Text: A lightweight and efficient text editor for writing and editing text as well as to code with speed and efficiency. During web application development the multi-line editing, syntax highlighting and plugin supporting make it very efficient.
- PhpStorm: Developed for PHP coding and intended for error identification and easy code refactoring processes.
- Visual Studio Code: Microsoft produces VS Code as a fast lightweight yet highly powerful code editor. Its features include support for multiple languages and Git integration and debugging functionality as well as available extensions.

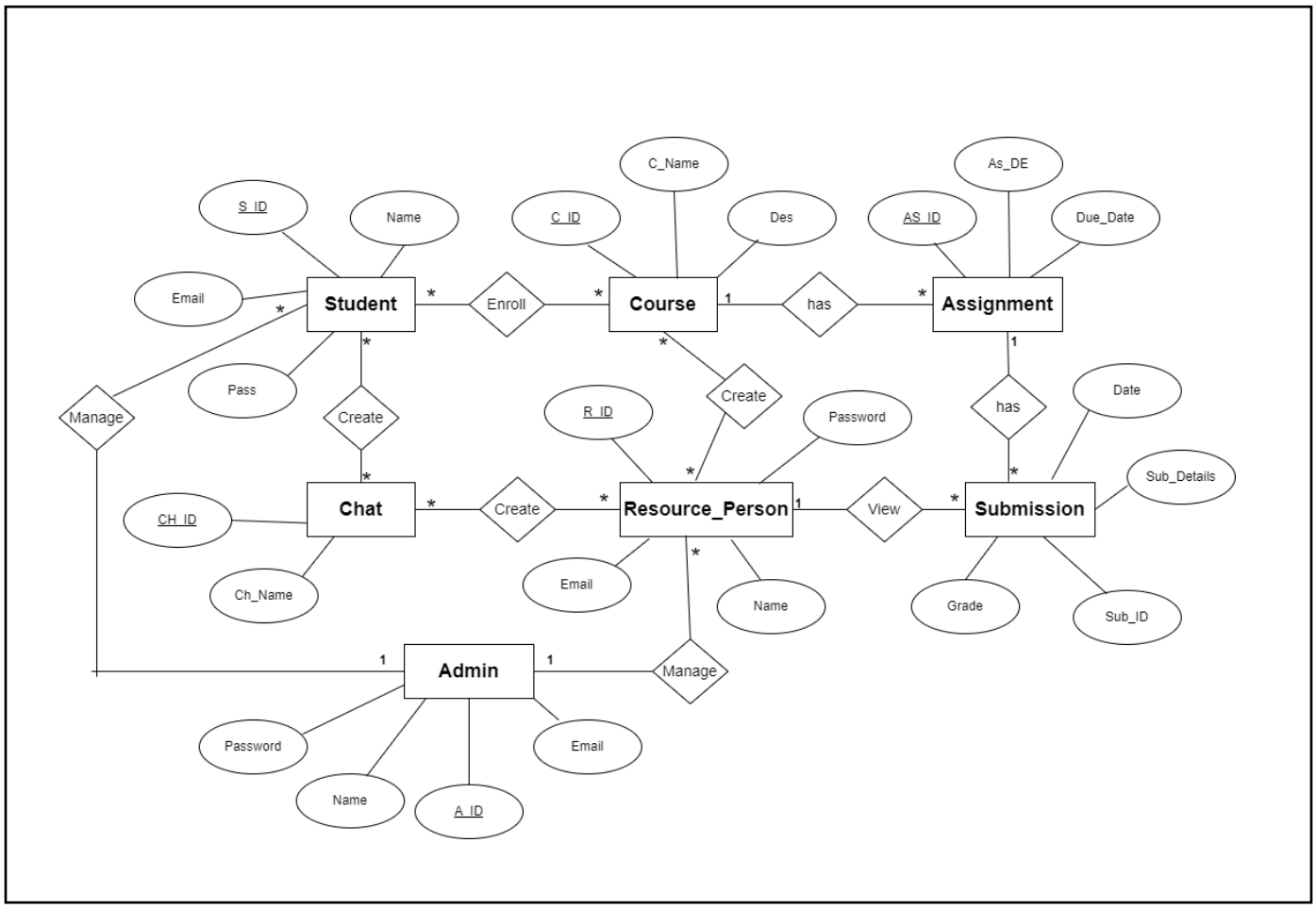
Chapter 3: Analysis

3.1 UML Diagrams

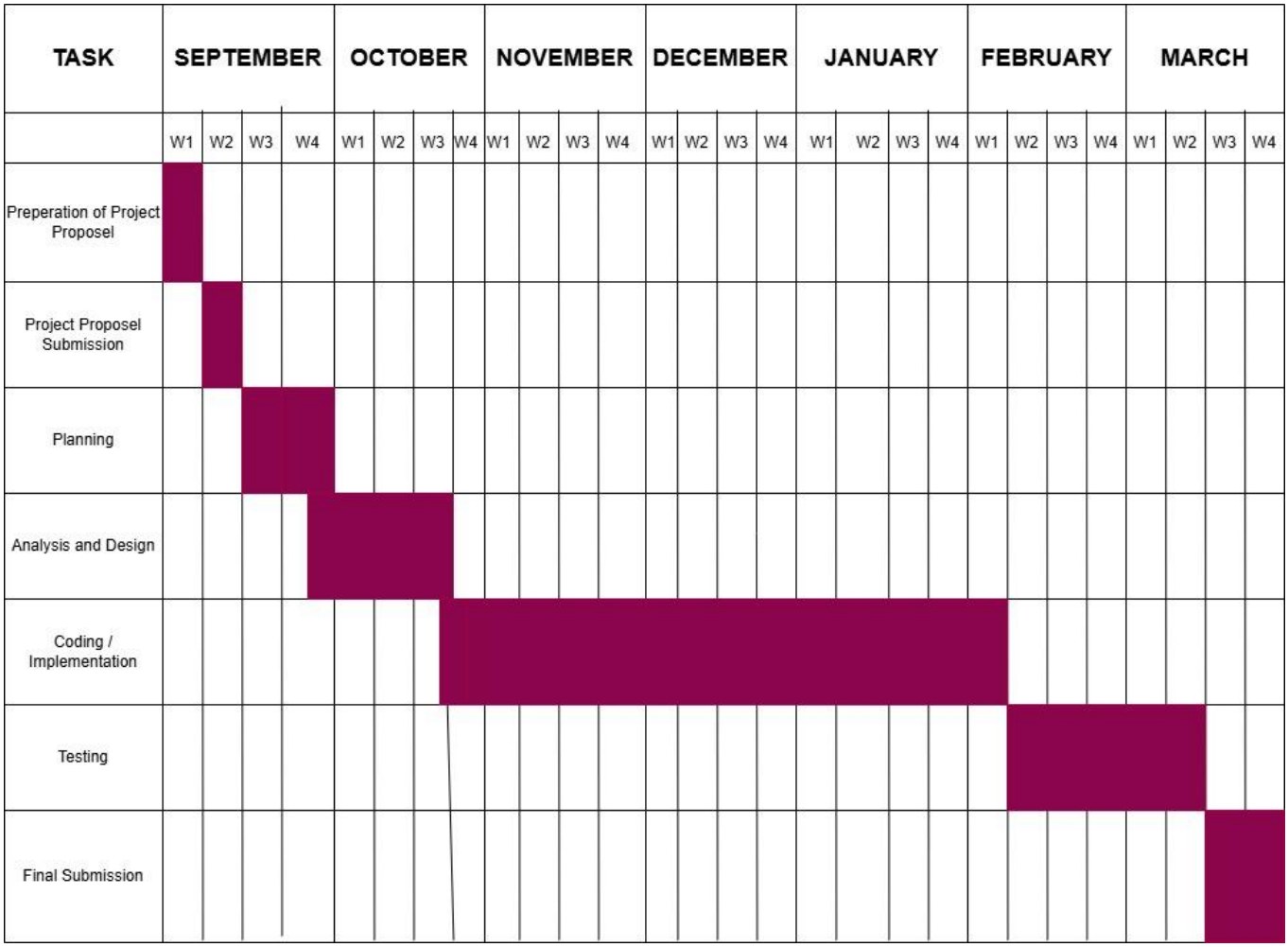
3.1.1 Use Case Diagram of Proposed System



3.1.2 ER Diagram for the proposed System



Chapter 4: Gantt Chart



Chapter 5: References

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Chapter 6: Appendices

6.1 Web Application Interfaces

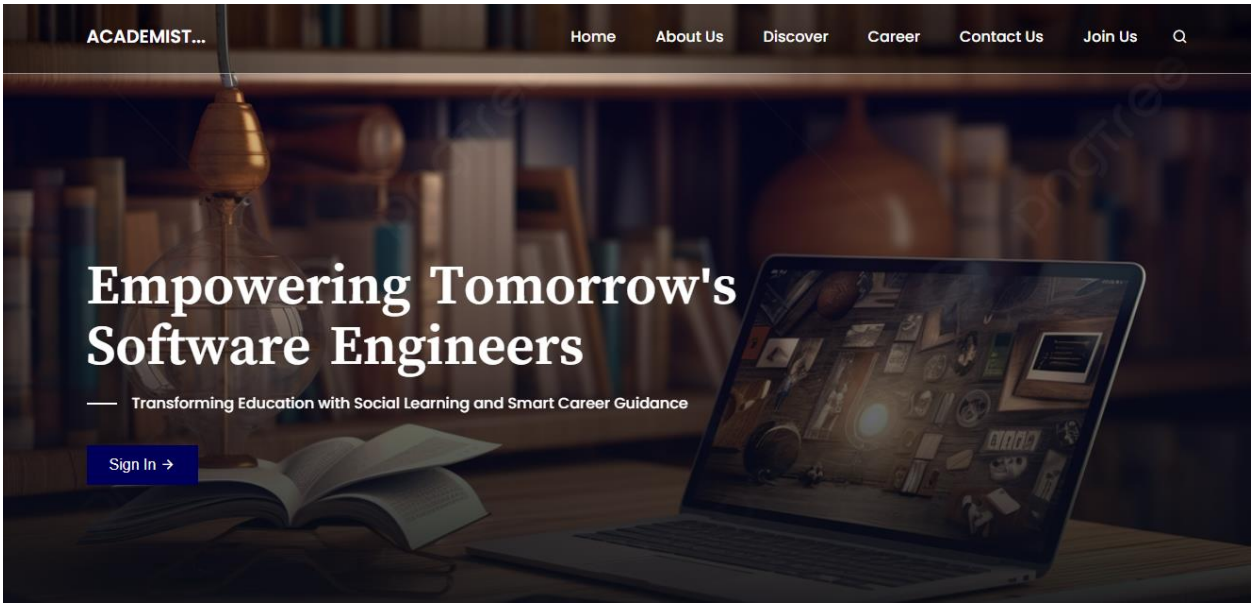


Figure 1 - Home_Page



Figure 2 - About_Us_Page

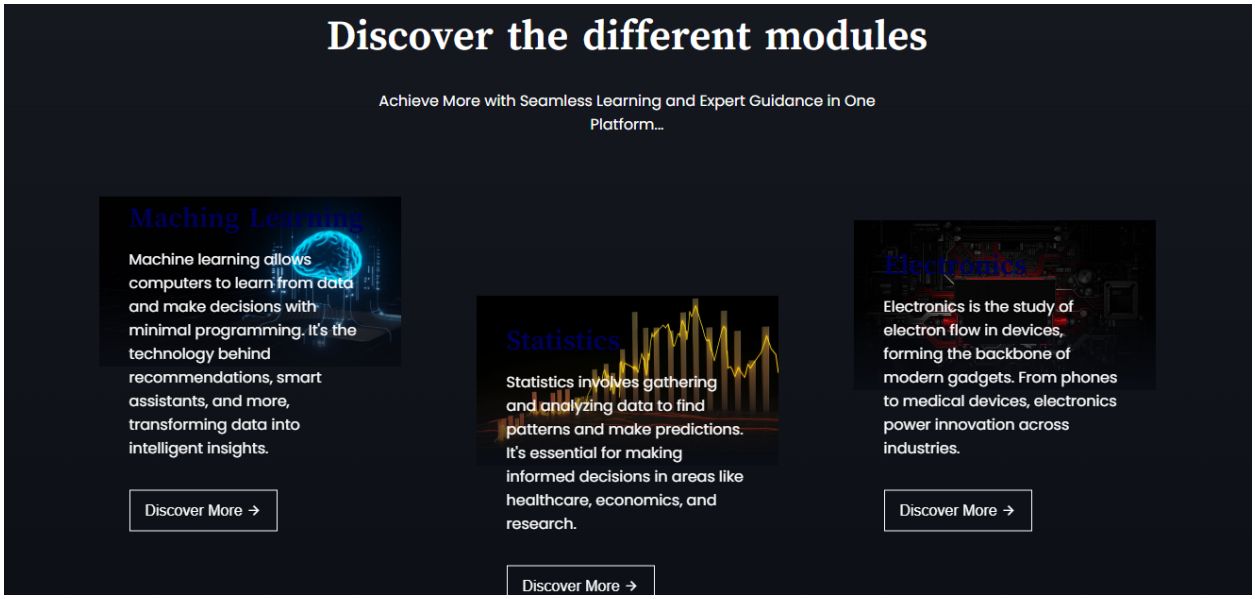


Figure 3 - Discover_Page_1

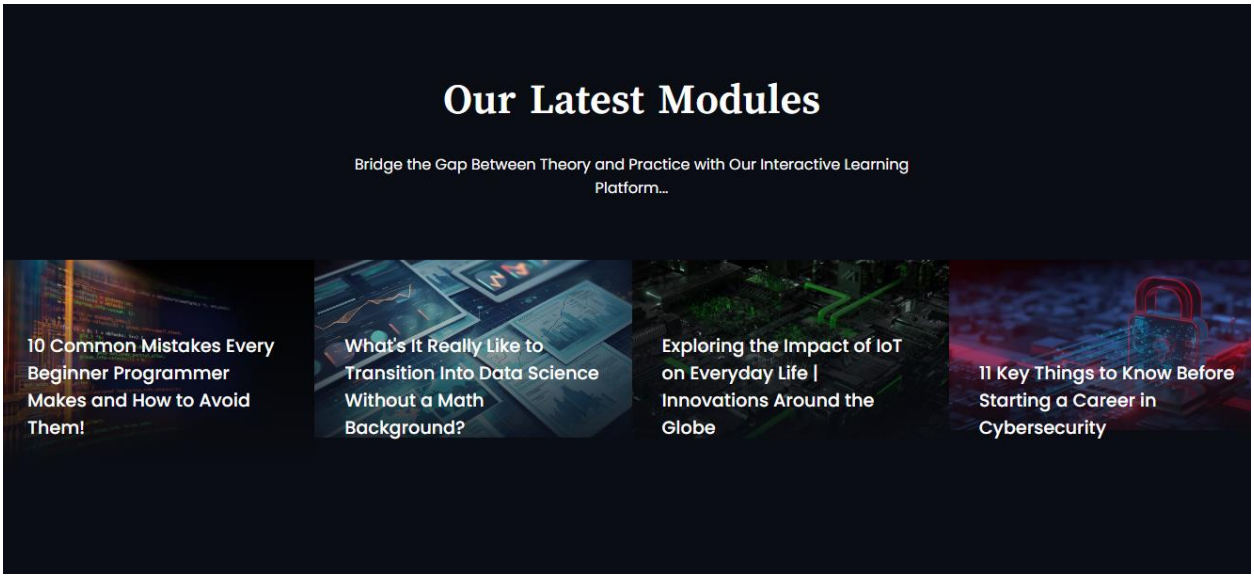


Figure 4 - Discover_Page_2



Figure 5 - Contact_Us_Page

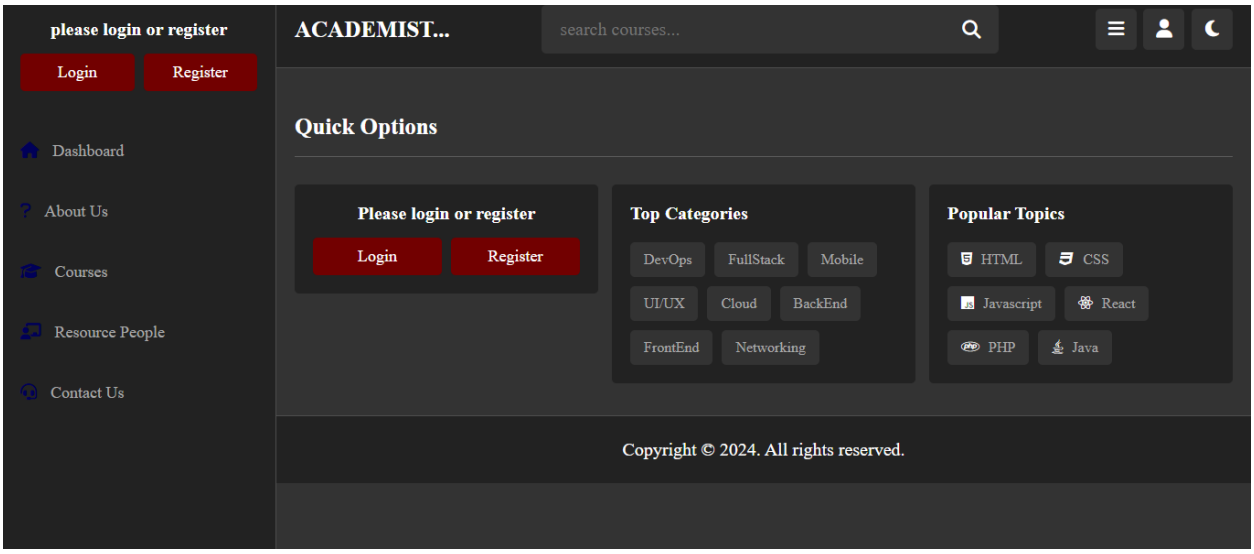


Figure 6 - Home_Page_1

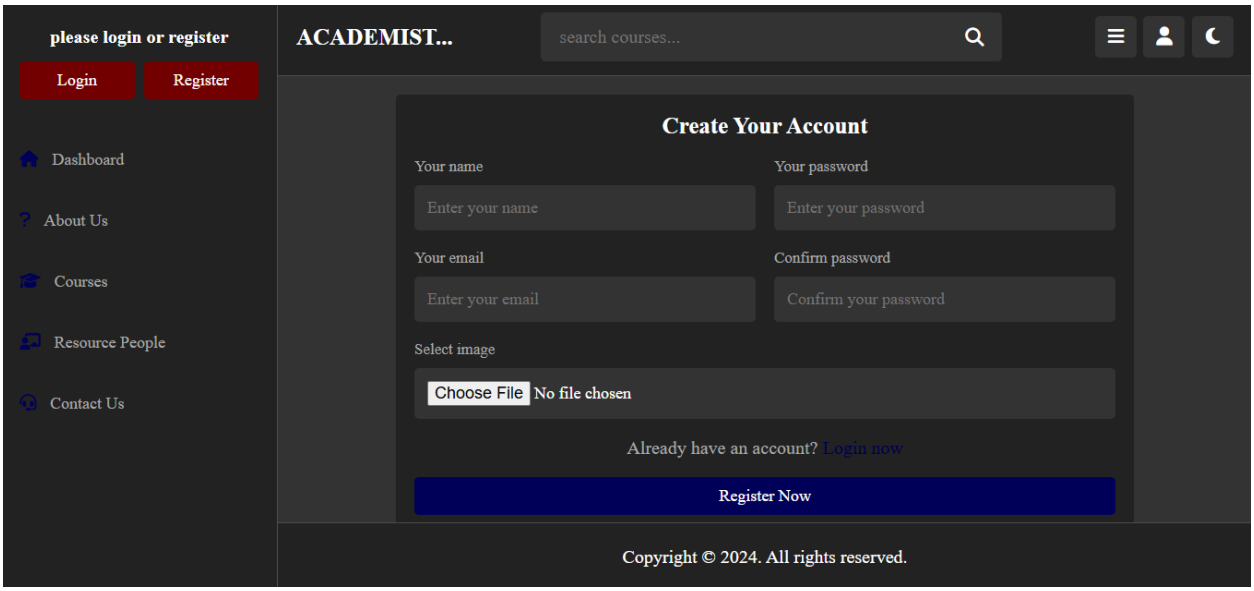


Figure 7 - Register_Page

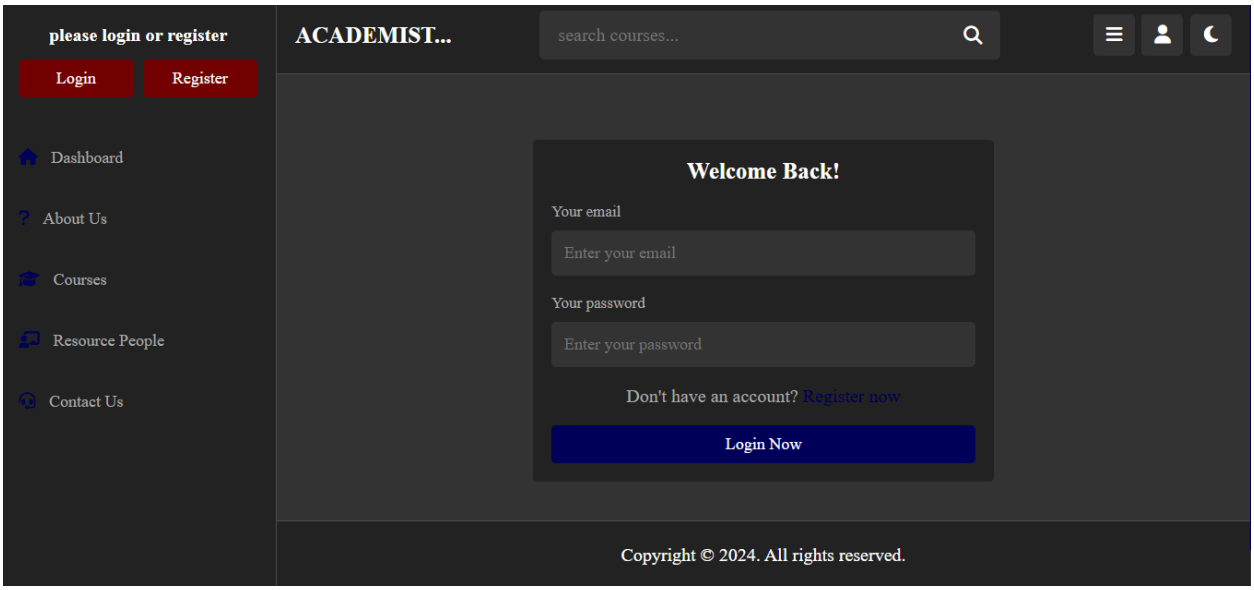


Figure 8 - Login_Page

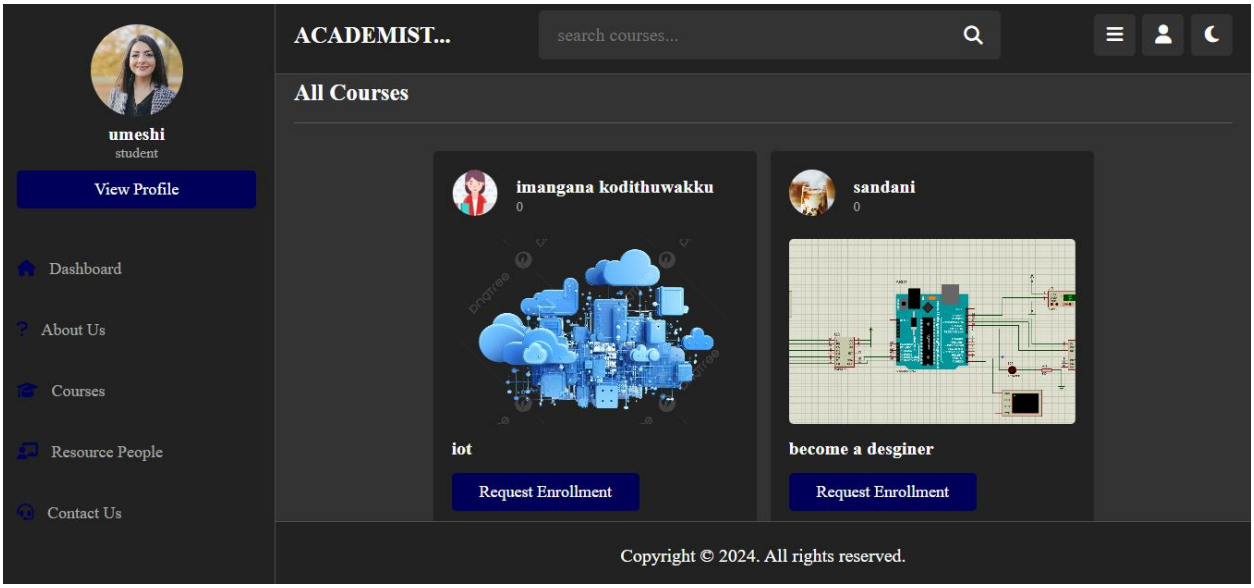


Figure 10 - Cources_Page

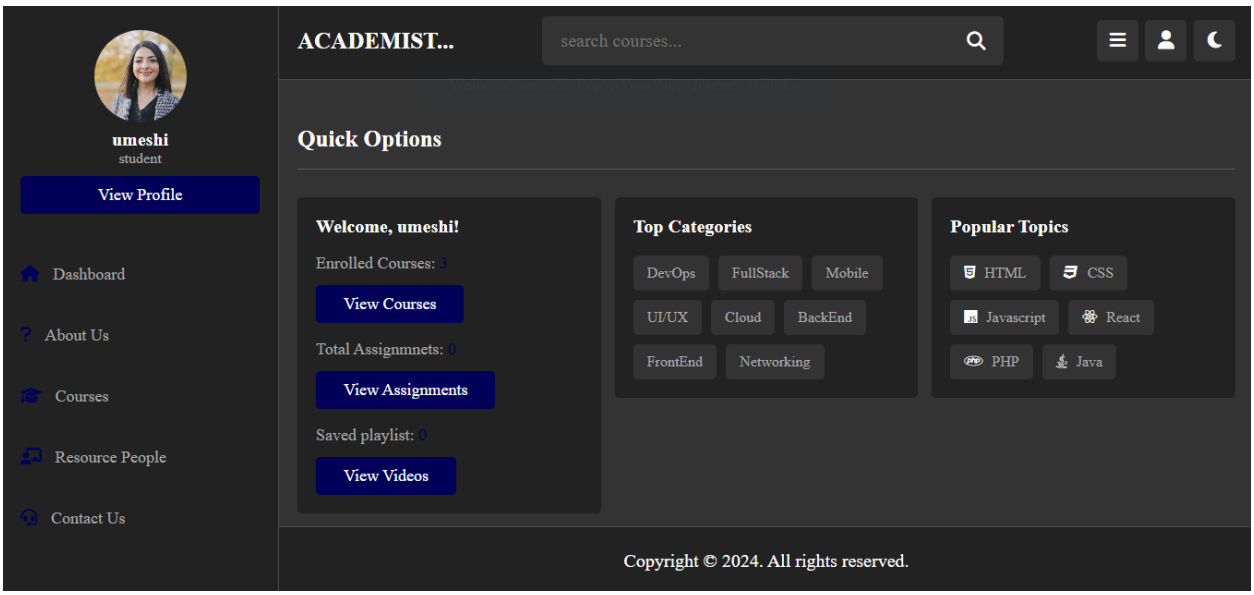


Figure 9 - Home_Page_2

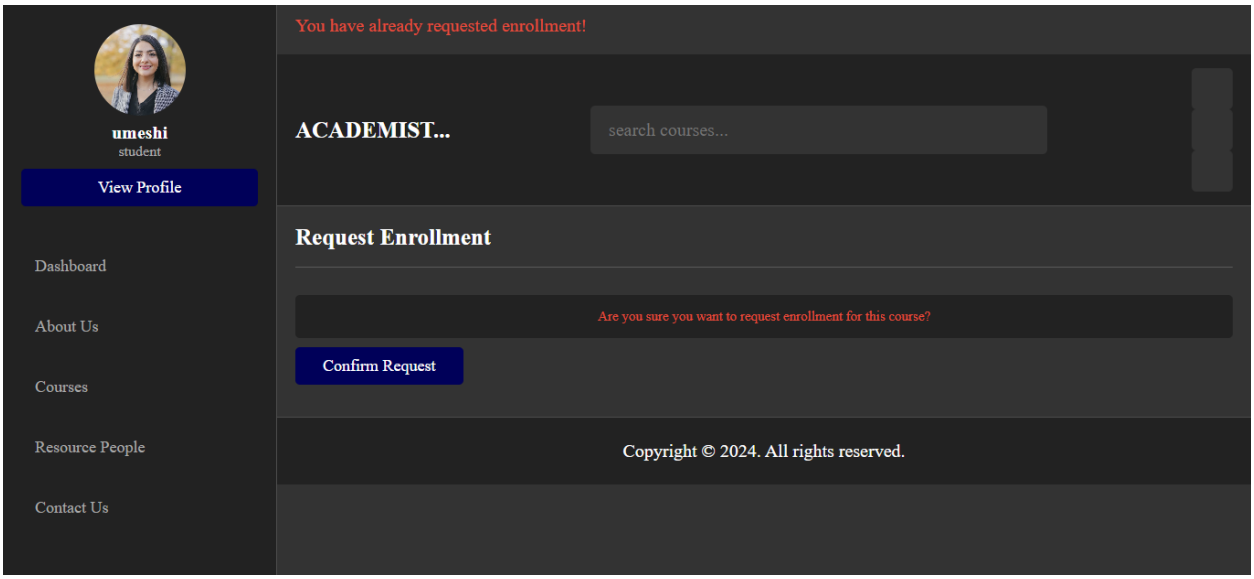


Figure 11 - Enrollment_Page

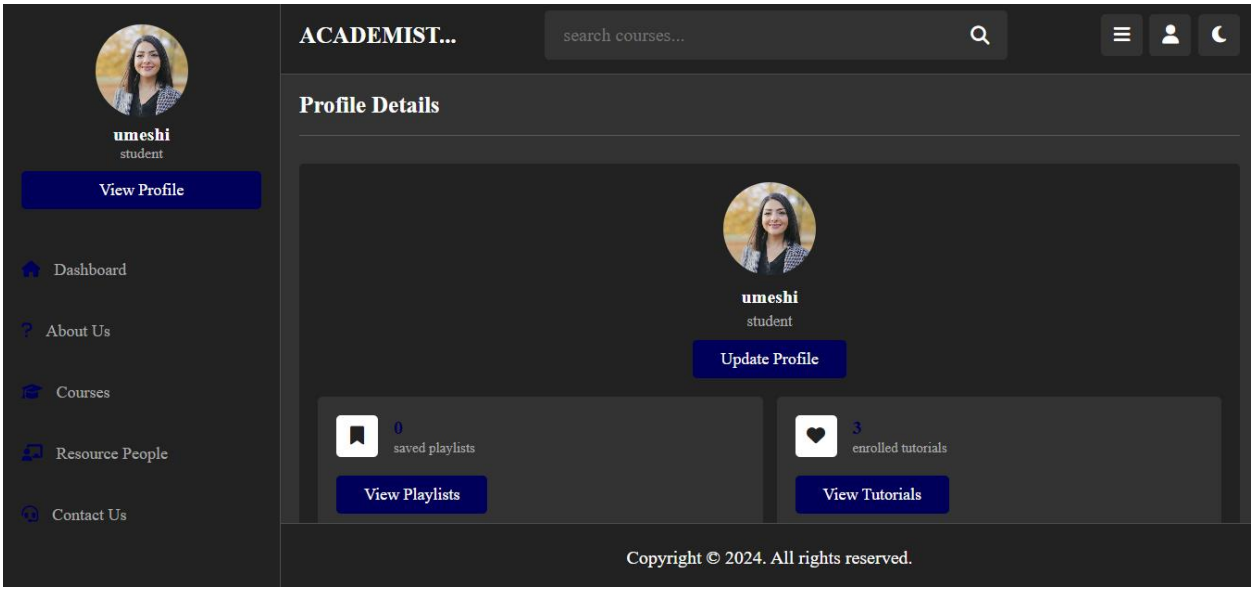


Figure 12 - Profile_Page

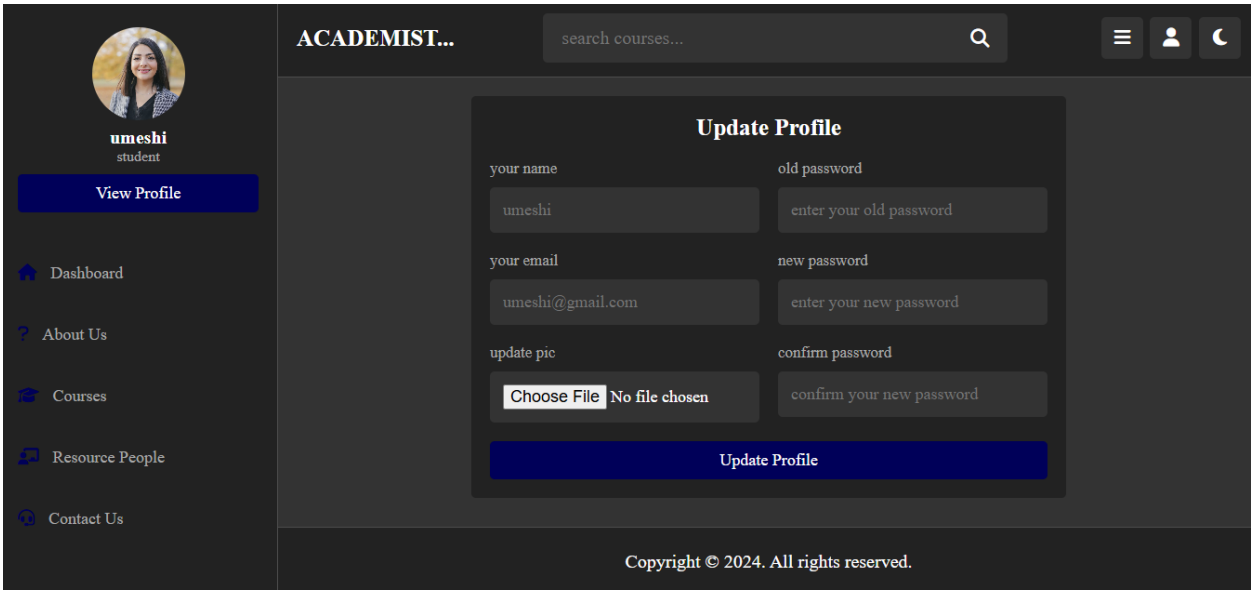


Figure 13 - Update_Profile_Page

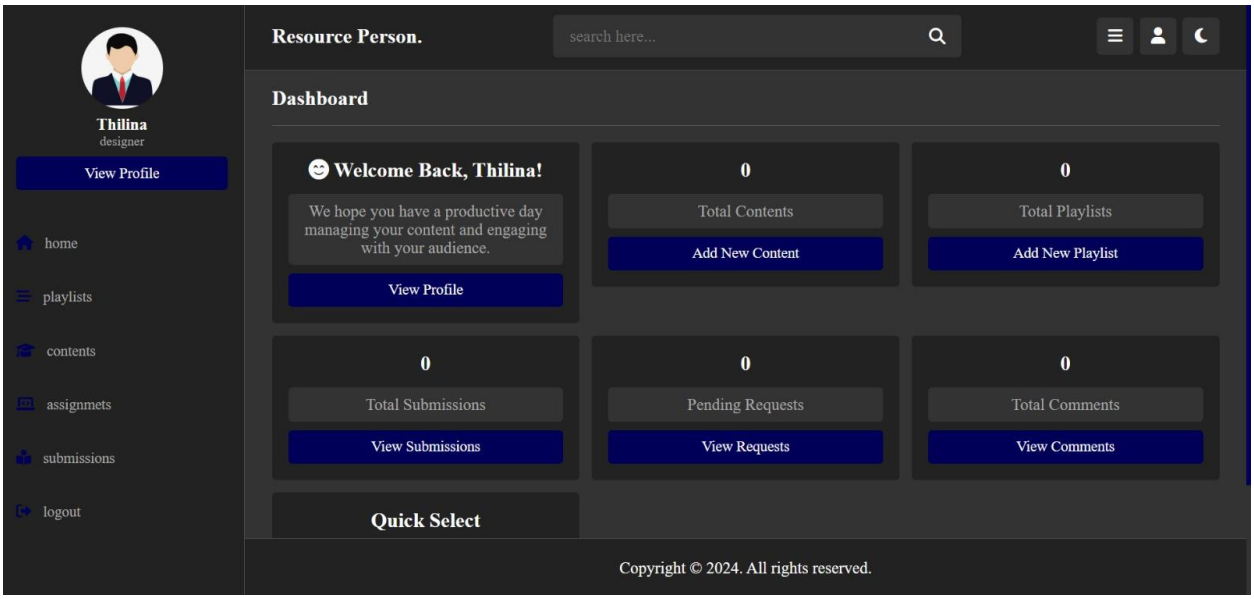


Figure 14 - Res_Peo_Dashboard

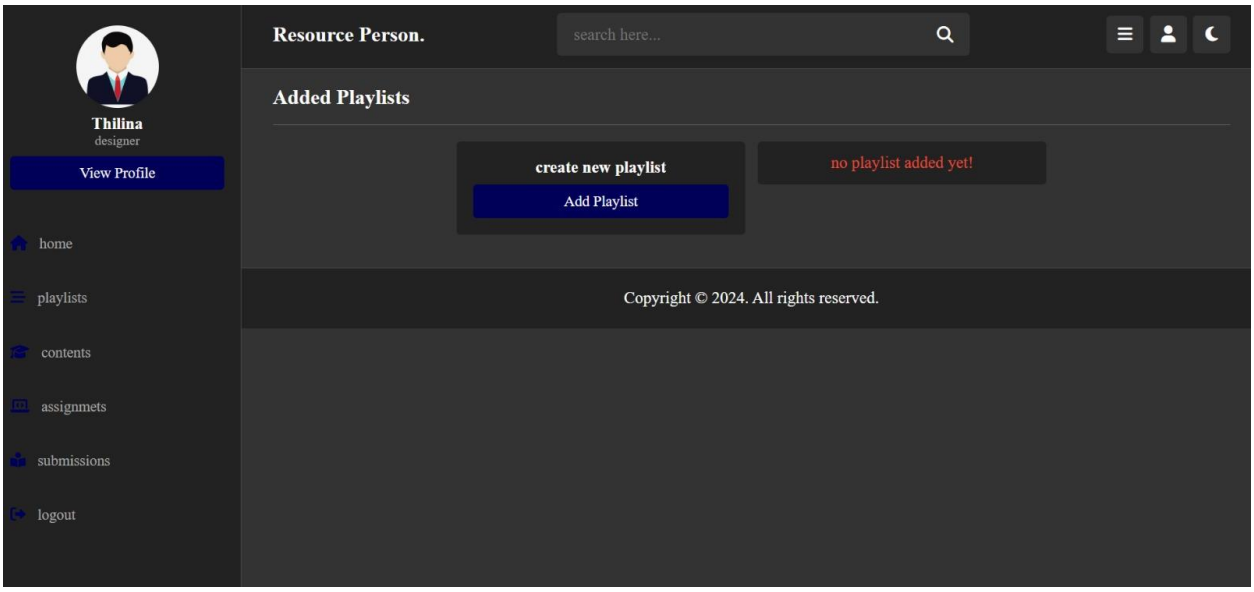


Figure 15 - Add_Content_Video_page

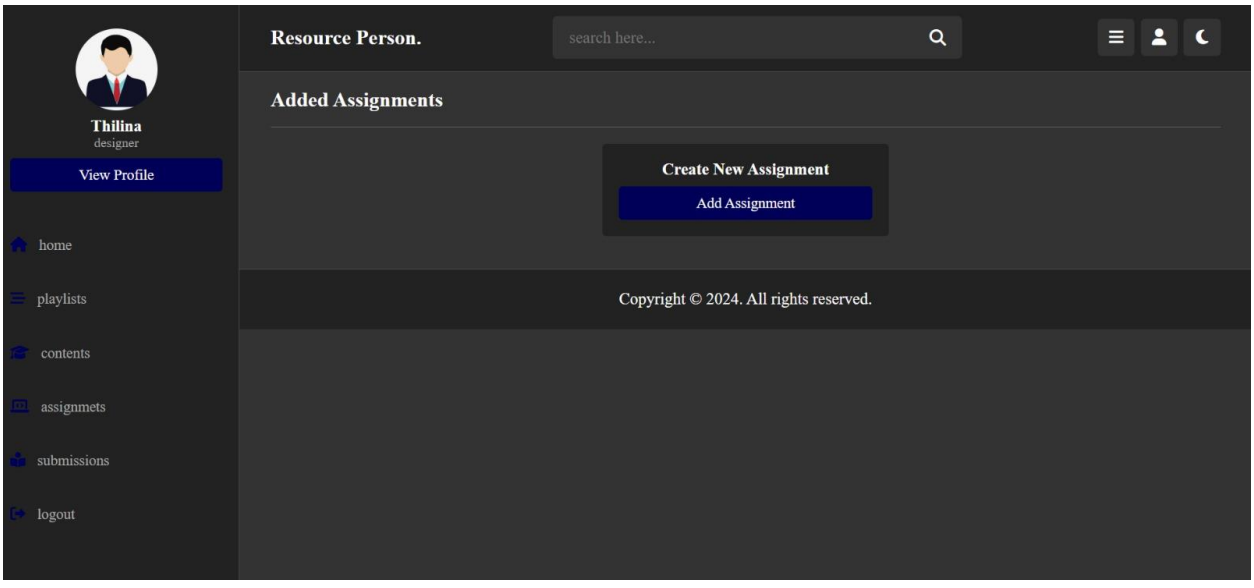


Figure 16 - Create_Assignment_Page

6.2 Mobile Application Interfaces



Figure 17 - Home_Page_Mobile

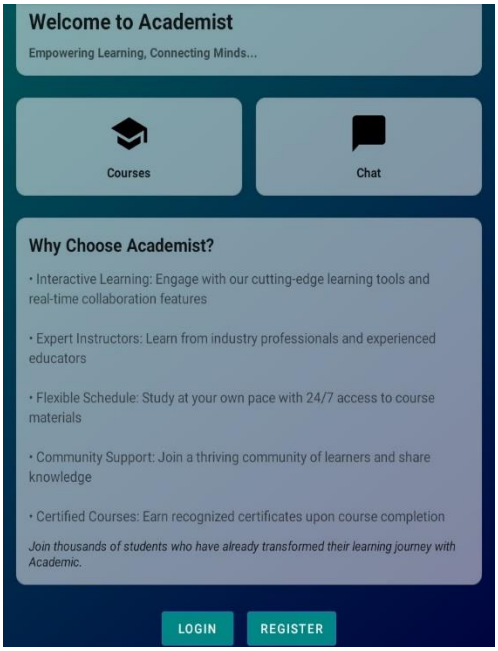


Figure 18 - Home_Page2

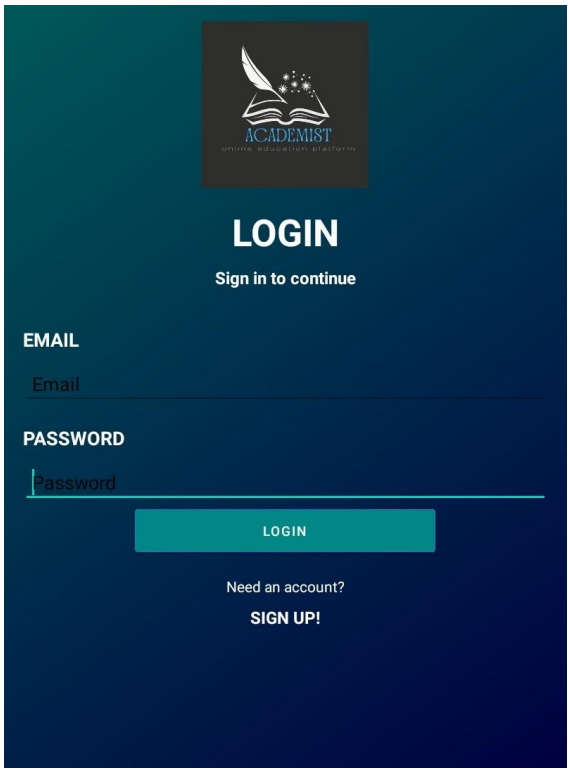


Figure 19 - Login_Page

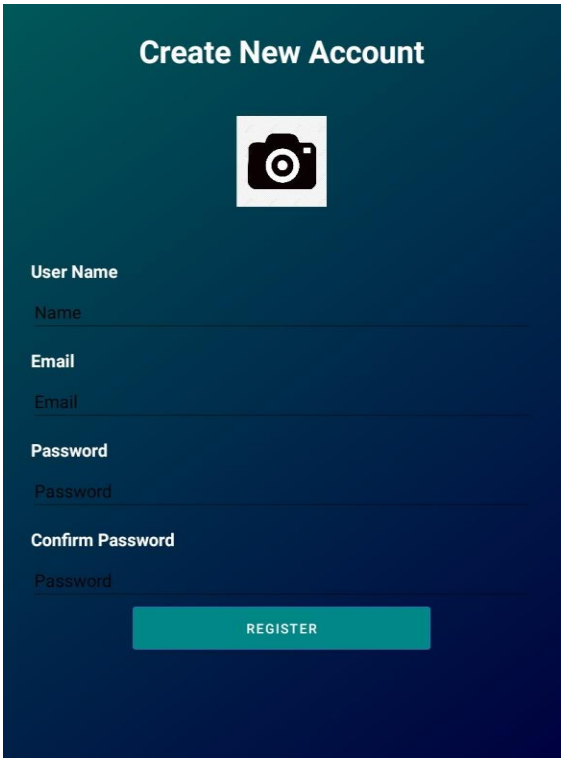


Figure 20 - Register_Page

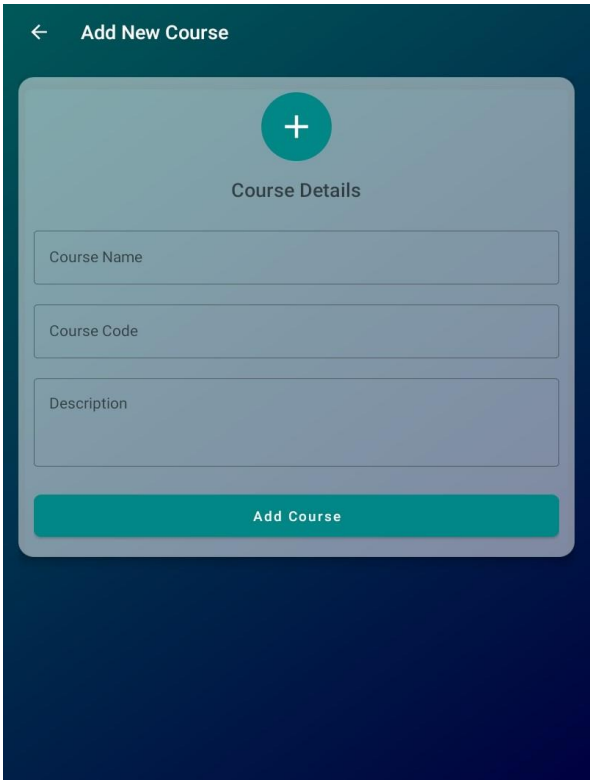


Figure 21 - Add_Cources_page

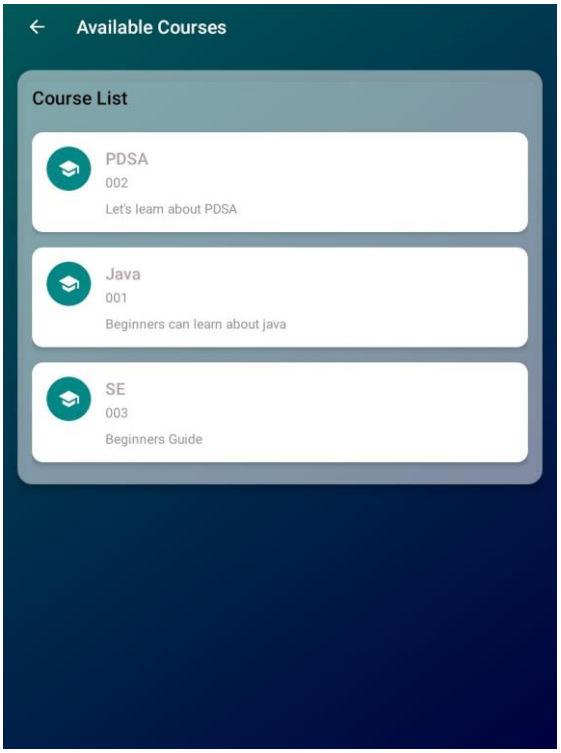


Figure 22 - Available_Cources_Page

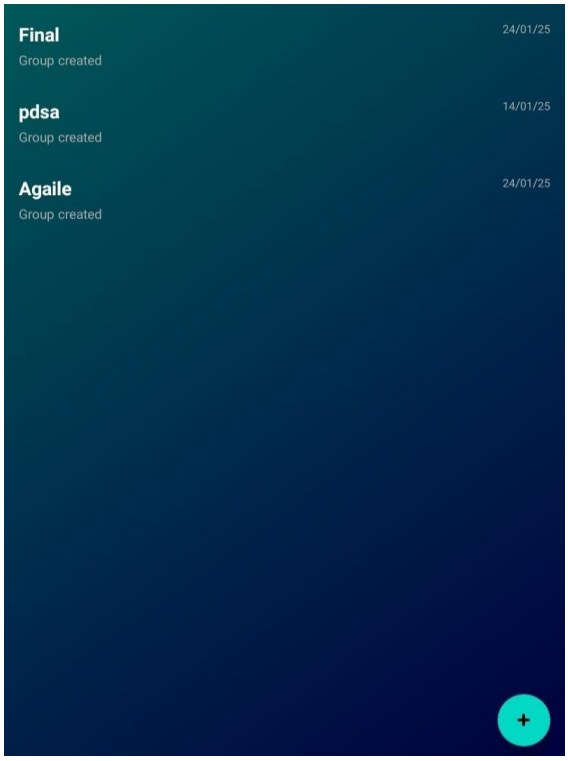


Figure 23 - Group-Page

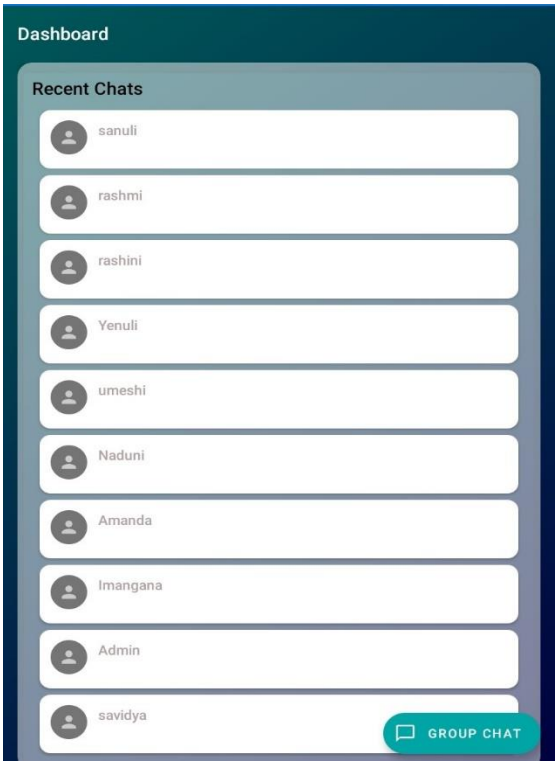


Figure 24 - Recent_Chat_page

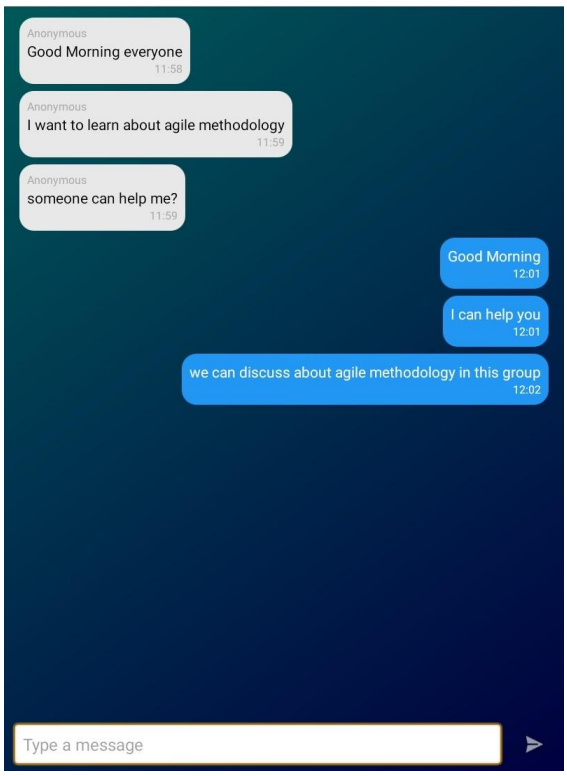


Figure 25 - Chat_Inbox_Page



Figure 26 - Feature_Page